

9.4.2.2.2 Scatter-based separability measure.

$$\begin{array}{lll} Q: & C_1 & (1,2) \quad (2,3) \quad (2,1) \\ & C_2 & (4,2) \quad (3,4) \quad (4,3) \end{array}$$

$$\text{Calculate (a) } J = \text{Tr}(S_w^{-1} S_B)$$

$$(b) J = \frac{\text{Tr}(S_B)}{\text{Tr}(S_w)}$$

$$\text{Solution } \textcircled{1} m_1 = \left(\frac{1+2+2}{3}, \frac{2+3+1}{3} \right)$$

$$= \left(\frac{5}{3}, 2 \right)$$

$$m_2 = \left(\frac{4+3+4}{3}, \frac{2+4+3}{3} \right)$$

$$= \left(\frac{11}{3}, 3 \right)$$

$$m = \left(\frac{1+2+2+4+3+4}{6}, \frac{2+3+1+2+4+3}{6} \right)$$

$$= \left(\frac{8}{3}, \frac{5}{2} \right)$$

$$m_1 - m_2 = \left(\frac{5}{3}, 2 \right) - \left(\frac{11}{3}, 3 \right) = (-2, -1)$$

$$\textcircled{2} S_B = (m_1 - m_2)(m_1 - m_2)^T \quad \text{from 4.8}$$

$$= \begin{pmatrix} -2 \\ -1 \end{pmatrix} \begin{pmatrix} -2 & -1 \end{pmatrix}$$

$$= \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

from 4.5

$$\textcircled{3} S_1 = \left(-\frac{2}{3}, 0\right)^T \left(-\frac{2}{3}, 0\right) + \left(\frac{1}{3}, 1\right)^T \left(\frac{1}{3}, 1\right) + \left(\frac{1}{3}, -1\right)^T \left(\frac{1}{3}, -1\right)$$

$$= \begin{bmatrix} \frac{4}{9} & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{9} & \frac{1}{3} \\ \frac{1}{3} & 1 \end{bmatrix} + \begin{bmatrix} \frac{1}{9} & -\frac{1}{3} \\ -\frac{1}{3} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{3} & 0 \\ 0 & 2 \end{bmatrix}$$

$$\textcircled{4} S_2 = \begin{pmatrix} \frac{1}{3} \\ -1 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & -1 \end{pmatrix} + \begin{pmatrix} -\frac{2}{3} \\ 1 \end{pmatrix} \begin{pmatrix} -\frac{2}{3} & 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{9} & -\frac{1}{3} \\ -\frac{1}{3} & 1 \end{pmatrix} + \begin{pmatrix} \frac{4}{9} & -\frac{2}{3} \\ -\frac{2}{3} & 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{9} & 0 \\ 0 & 0 \end{pmatrix}$$

$$= \begin{bmatrix} \frac{2}{3} & -1 \\ -1 & 2 \end{bmatrix}$$

$$\textcircled{5} S_W = S_1 + S_2$$

$$= \begin{bmatrix} \frac{4}{3} & -1 \\ -1 & 4 \end{bmatrix}$$

$$\textcircled{b} \text{Tr}(S_w) = \frac{16}{3}$$

$$\begin{aligned} \text{(a)} \quad J &= \text{Tr}(S_w^{-1} S_B) \\ &= \text{Tr}\left(\begin{bmatrix} \frac{4}{3} & -1 \\ -1 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}\right) \\ &= \text{Tr}\left[\begin{array}{cc} \frac{54}{13} & \frac{27}{13} \\ \frac{20}{13} & \frac{10}{13} \end{array}\right] \\ &= \frac{64}{13} \\ &= 4.9231 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad J &= \frac{\text{Tr}(S_B)}{\text{Tr}(S_w)} = \frac{5}{\frac{16}{3}} \\ &= \frac{15}{16} \\ &= 0.9375 \end{aligned}$$